High-Fidelity Art Restoration Utilizing Latent Diffusion

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Abstract. Art restoration is a meticulous, time-intensive process performed by professional museum conservators, often requiring weeks or months of research and manual work. Generative image restoration techniques can significantly shorten this process; however, previous methods have struggled to restore intricate artworks accurately while preserving their artistic style and appearance. Prior research aiming to improve art restoration using a VAE or pixel-space diffusion methods has struggled to capture long-range dependencies and maintain stylistic consistency in their reconstructions. This paper proposes a novel latent diffusion approach for artwork restoration that employs a residual variational autoencoder (Res-VAE) and an image-conditioned diffusion transformer (DiT), leveraging both the compressed latent representation for computational efficiency and cross-attention blocks for improved restoration fidelity. This method is evaluated on a comprehensive dataset spanning a wide range of artistic styles and compared against state-of-the-art restoration models. Experimental results show that the model outperforms these methods both qualitatively and quantitatively in repairing damaged artwork. This approach advances cultural preservation by enabling more accurate and efficient digital restoration of artwork across diverse artistic styles.

Keywords: Generative models · Artwork restoration · Latent diffusion · Masked image modeling · Diffusion transformer

1 Introduction

Artwork has been integral to the cultural and historical fabric of society throughout history. It serves as a visual record of the changing ideologies and values across time, but as these works age, they inevitably suffer from damage. Environmental factors, accidental damage, and paint chipping require expert art conservators to perform a time-consuming and expensive restoration process. Digital art restoration serves as a means to reduce the complexity and time commitment of this process, while still maintaining the vision and creativity of the original artist.

Despite recent developments in generative AI that enable high-quality image generation across a wide range of areas, general-purpose image generation

models such as DALL-E and Midjourney are fundamentally unable to achieve artifact-free restoration on artwork. This limitation arises because these models are primarily trained on billions of natural images, which typically contain gradual color transitions and well-defined structures [1]. In contrast, artwork is often highly abstract and follows non-generalizable semantics, making it difficult for natural-image-trained models to preserve both stylistic and structural aspects during restoration. In one instance, VanceAI's Photo Restorer failed to restore the infamous *Ecce Homo* painting, only filling in small cracks and refining details already present rather than reconstructing the original artistic intent [19].

In the last few years, numerous models have been developed for art restoration tasks using a variety of methods. Models using Variational Autoencoders (VAEs) [10] and Generative Adversarial Networks (GANs) [11] have been able to achieve significant results by learning the local image structure, but often fail to restore large areas. Diffusion models such as Denoising Diffusion Probabilistic Models (DDPMs) presented by Ho et al. [7] and Diffusion Transformers (DiTs) presented by Peebles and Xie [14] utilize attention mechanisms [20] to capture long-range dependencies and global image context, producing high-quality image restorations. Nevertheless, many current diffusion approaches face limitations in adapting to the stylistic nuances of artwork, reducing their effectiveness across diverse art styles and eras.

To overcome these limitations, I propose a novel ensemble latent diffusion model incorporating a modified DiT that operates within the latent space of a residual VAE (Res-VAE) proposed by Ditria and Drummond [5]. This design leverages the strengths of each architecture specifically for artwork restoration tasks. Both the VAE's ability to encode complex structures into a compact latent space and the DiT's ability to gradually refine the latent representation allow for a detailed restoration.

In summary, the main contribution of this paper is twofold: (1) I propose a novel latent diffusion model that captures both the low-level structure and high-level stylistic details of the artwork to produce high-fidelity artwork restoration without additional context; and (2) I conduct extensive experiments against other models as well as an ablation study on a diverse dataset to validate the accuracy and adaptability of my model.

2 Related Work

The field of image restoration has evolved rapidly over the past decade, progressing from convolutional architectures to diffusion-based models. In the absence of prompts or class conditioning, vanilla inpainting models must learn a broad image distribution and infer missing content solely from surrounding pixels. Early work by Yeh et al. [24] proposed combining a VAE with a Generative Adversarial Network (GAN) to search for the closest encoding of corrupted images for filling in the missing regions. The introduction of iterative refinement through DDPMs [7] and attention mechanisms [20] marked a significant shift in image inpainting as they allow the model to incorporate global semantic structure across

the entire image. Lugmayr et al. [13] proposed an unconditional DDPM that enforces conditioning only during sampling via resampling known regions, at the cost of increased inference time. To address this, Corneanu et al. [4] introduced LatentPaint, which performs diffusion in a VAE's compressed latent space to reduce computational overhead. However, artwork restoration presents fundamentally different challenges compared to general inpainting due to its sparse, high-contrast feature distributions.

Therefore, digital artwork restoration demands models that extend beyond smooth, generic priors learned from natural images to accurately preserve these complex feature distributions. Prior work addressing this challenge can be broadly categorized into CNN/VAE methods and diffusion methods.

CNN and VAE-based approaches have proven to be effective in capturing spatial hierarchies and learning compressed representations of the artwork distribution. Gupta et al. [6] proposed a two-stage hybrid model consisting of a mask generator R-CNN and partial convolution U-Net for inpainting. Sun et al. [18] proposed a triplet-domain translation network with two VAEs to create mappings between damaged and clean image domains through a shared latent space. Sankar et al. [2] proposed a distributed denoising CNN model that addresses various distortion types at multiple degradation levels. Peng et al. [15] proposed using dual-domain partial convolution operating in both spatial and frequency domains to handle irregular holes. Zhang et al. [26] proposed decomposing the restoration into subtasks (luminance, sketch, and color reasoning) and using attention mechanisms to fuse them together. Phadke et al. [16] proposed combining a dual VAE architecture and GAN-based mapping network to align degraded and clean images in a shared latent space while preserving facial features. However, these implementations struggle with large-scale missing regions and often produce blurred textures due to their inability to capture long-range dependencies necessary for coherent reconstruction.

Diffusion-based approaches have been shown to be effective in generating fine artistic reconstructions with global consistency due to their iterative denoising processes. Zhao et al. [27] proposed using a diffusion-based heterogeneous network and dual channel attention blocks to both restore and refine large-scale image patches. Wang et al. [22] proposed a zero-shot framework using a pretrained DDPM-like architecture to restore only the masked regions without artwork-specific training. Wang et al. [21] proposed combining a denoising diffusion probabilistic model with the Vision Transformer (ViT) architecture, utilizing long skip connections to fuse global and local features. Jia et al. [9] proposed a structure and texture-guided diffusion network with attention-based feature fusion to balance structural and textural integrity. However, these diffusion-based methods face challenges in computational efficiency due to operating in pixel space and fail to generalize across diverse artistic styles due to their limited training data.

With these architectures and limitations in mind, I will present a novel ensemble model that performs diffusion in the latent space, reducing computational

complexity while preserving the structure and texture necessary for creating a high-quality restoration.

3 Method

In this section, I present my art restoration model that incorporates a ResVAE [5] and modified DiT [14] to reconstruct damaged artwork. I will elaborate on the necessary background, masking process, model architecture, loss function, and training procedure.

3.1 Preliminaries

Variational Autoencoders Variational autoencoders (VAEs) [10] are a class of unsupervised generative models that learn to encode images into a probabilistic latent space ($\mathbf{z}$). This latent space is constrained to follow a prior distribution $p(z)$, typically a standard Gaussian distribution $N(0, 1)$, allowing the model to sample this distribution and generate new images. These models are tasked with optimizing the Evidence Lower Bound (ELBO), ensuring that both the reconstruction quality and latent space regularization (Kullback-Leibler divergence) allow the model to produce the highest quality results.

$$\mathcal{L}_{\text{VAE}}(\mathbf{x}) = \underbrace{\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z})]}_{\text{Reconstruction Log-Likelihood}} - \underbrace{D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x})\|p(\mathbf{z}))}_{\text{KL Divergence}} \tag{1}$$

where D_{KL} denotes the Kullback-Leibler divergence function. VAEs are often constructed in a U-Net [17] that includes both an encoder and a decoder. The encoder network ($q_\phi(\mathbf{z}|\mathbf{x})$) maps the input to a latent representation, while the decoder network ($p_\theta(\mathbf{x} \mid \mathbf{z})$) takes this latent representation and tries to reconstruct the original image from it. The latent space learned by the VAE serves as a compressed representation of the input data distribution to efficiently generate samples; however, this compression limits the complexity that the VAE can generate. Residual skip connections between encoder and decoder layers are used to partially compensate for this problem by allowing high-dimensional information to be directly preserved after the latent space bottleneck. Nevertheless, the fundamental diversity of artwork leads to inefficient latent representations within VAEs that either over-allocate capacity to uniform regions or under-represent critical artistic details.

Diffusion Transformers Diffusion models are another class of generative models that can produce high-resolution images through their forward and reverse diffusion processes. By learning how to iteratively reverse a noise-filled image, they can generate new, high-quality images from random noise. The forward diffusion process gradually adds Gaussian noise to the input image (z_0) across T timesteps using a Markov chain, eventually transforming the image into pure noise (z_t) through this equation:

$$q(z_t|z_{t-1}) = \mathcal{N}(z_t; \sqrt{1-\beta_t}z_{t-1}, \beta_t I) \tag{2}$$

where $q(z_t|z_{t-1})$ denotes the forward diffusion distribution, t represents the timestep, and β_t is a variance parameter that defines the noise schedule. Once the image has been turned into pure noise, the reverse inference process is tasked with predicting how much noise was added at each timestep, such that it can be removed. This reverse process is defined as:

$$p_\theta(\mathbf{z}_{t-1}|\mathbf{z}_t) = \mathcal{N}(\mathbf{z}_{t-1}; \mu_\theta(\mathbf{z}_t, t), \Sigma_\theta(\mathbf{z}_t, t)) \tag{3}$$

where $p_\theta(z_{t-1}|z_t)$ denotes the reverse diffusion distribution parameterized by θ, and μ_θ and Σ_θ are the mean and variance functions conditioned on the noisy image z_t and timestep t. The architecture used for the reverse process significantly impacts the model's denoising capabilities. Foundational works such as DDPMs [7] use a U-Net with convolutional layers and sparse self-attention blocks. However, even with limited attention, their predominantly convolutional backbone constrains its ability to capture long-range dependencies and global context. Transformer-based diffusion models, such as ViTs [3], address this limitation by representing the entire latent representation as a flat sequence of patches and applying attention across all patches at each timestep, enabling every patch's denoising to be informed by global context. DiTs [14] build on this idea but replace the U-Net with a transformer architecture and typically operate in the latent space rather than the pixel space, reducing the computational complexity and improving the generation quality across the image.

For artwork restoration, this combination of latent-space efficiency and transformer-based global attention is extremely valuable. The iterative refinement allows the model to gradually recover fine brushstrokes and textures while preserving artistic style and composition.

3.2 Proposed Model

Figure 1 shows the architecture of the proposed model. The input to the model is the masked image and the output is a reconstructed image where the masked regions are filled with the model's inferences. In order to artificially damage the artwork, masked image modeling (MIM) was applied across the 128×128 image, randomly masking 16×16 patches with a 40% probability to mimic missing regions and tears. The model integrates both a Res-VAE [5] with skip connections and a modified DiT that replaces its standard class-conditioning with image-conditioning on the intact regions of the latent representation. The Res-VAE compresses artwork into a computationally efficient latent space while its residual skip connections preserve the fine-grained details essential for authentic restoration. The DiT operates within the Res-VAE's latent space, conditioning on the intact regions of the latent representation to incorporate structural and semantic cues while denoising the noisy latent, thereby maintaining artistic consistency during inference.

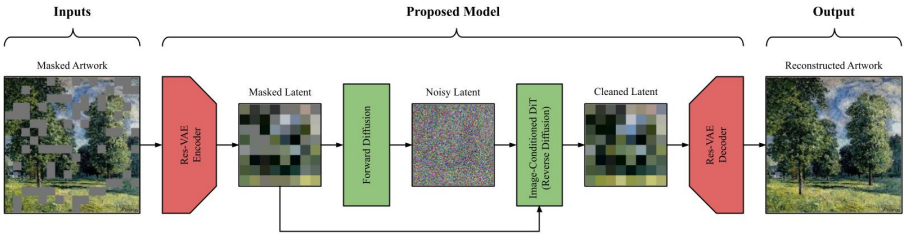


Fig. 1. Overview of the proposed model's architecture. The masked image is fed through the proposed model architecture to produce a reconstruction of the missing areas.

The Res-VAE's encoder first processes the masked image through an initial convolution, followed by a series of four *ResDown* blocks that progressively downsample the image while increasing its channel depth. To ensure that high-level data is preserved, a *ResBlock* is applied before the features are transformed into the latent distribution parameters, from which z is sampled via the reparameterization trick. The decoder mirrors this architecture, upsampling the latent representation through another *ResBlock* and four *ResUp* blocks. Both the internal residual connections within each type of block and the skip connection between the encoder and decoder preserve the fine-grained details that would otherwise be lost. Ultimately, the output is put through a tanh activation function to regularize the reconstruction output.

The DiT serves as the core restoration mechanism and operates entirely within the Res-VAE's latent space. The forward diffusion process begins by fully corrupting the latent representation z from the VAE, adding Gaussian noise across 1000 timesteps (T) using the previously defined Markov Chain (2). The noisy latent is then converted into a sequence of 1D tokens through the Patchify operation, while the original latent of the masked image z is concatenated with the timestep embedding $\tau(t)$ to form the conditioning signal:

$$c = \text{Patchify}(z) \oplus \tau(t) \tag{4}$$

Once this is complete, the reverse diffusion process begins, with both the noisy latent and conditioning embeddings fed into a series of 12 DiT blocks, where they are integrated through each block's multi-head cross-attention mechanism. This cross-attention allows the model to align the noisy latent with the conditional guidance, effectively leveraging the available image context and maintaining global consistency. After the inputs are processed through all 12 transformer blocks, a final linear projection and unpatchify operation reconstructs the complete denoised latent representation from the token sequence.

While VAEs and diffusion models have individually been applied to image restoration, artwork restoration poses unique challenges due to sparse feature

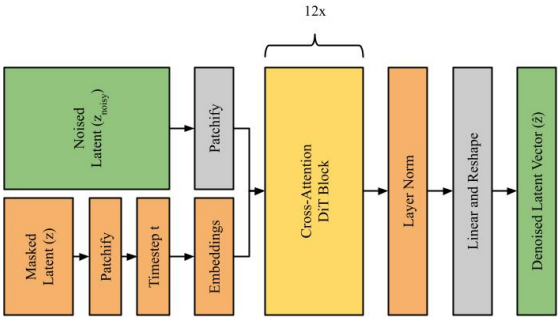


Fig. 2. Overview of the modified DiT architecture using the masked latent and timestep as the embeddings

distributions, diverse artistic styles, and the need to maintain authenticity. To address this, my proposed model employs a modified image-conditioned DiT approach, which preserves both restoration quality and artistic intent while reducing the computational complexity required.

3.3 Loss Function

In this section, I present the loss functions used to train the proposed model. The training process was conducted in two stages, each requiring a distinct loss function: (1) Res-VAE training; (2) DiT Training. I will elaborate on components of each loss function and explain the rationale behind their selection.

VAE Training Loss The VAE training loss combines the standard KL divergence term (5) to regularize the latent distribution toward a Gaussian prior and mean squared error (MSE) (6) for pixel-level accuracy:

$$\mathcal{L}_{\text{KL}} = D_{\text{KL}}(q_\phi(z|x)\|p(z)) \tag{5}$$

$$\mathcal{L}_{\text{MSE}} = \|x - \hat{x}\|^2 \tag{6}$$

where x represents the original unmasked image, $\hat{x}$ is the reconstructed output, z denotes the latent representation, $q_\phi(z|x)$ is the approximate posterior distribution created by the encoder, and $p(z)$ is the prior distribution. However, pixel-level metrics alone lead to blurry reconstructions that lose artistic texture. To preserve high-level characteristics such as texture and structure, I incorporate Learned Perceptual Image Patch Similarity (LPIPS) loss [25], specifically using the VGG19 model.

Nevertheless, LPIPS compares images based on intermediate neural network activations, which capture general textures and patterns but often miss fine-grained details such as individual brushstrokes and sharp edges. To preserve

these high-frequency components, I incorporate Fourier Loss (7), which penalizes differences in the frequency domain:

$$\mathcal{L}_{\text{Fourier}} = \|\mathcal{F}(x) - \mathcal{F}(\hat{x})\|^2 \tag{7}$$

where $\mathcal{F}$ denotes the Fourier transform. The complete VAE training objective combines all components with hyperparameters β_{KL}, β_{MSE}, β_{LPIPS}, and β_{Fourier} controlling their relative weighting:

$$\mathcal{L}_{\text{VAE}} = \beta_{\text{KL}}\mathcal{L}_{\text{KL}} + \beta_{\text{MSE}}\mathcal{L}_{\text{MSE}} + \beta_{\text{LPIPS}}\mathcal{L}_{\text{LPIPS}} + \beta_{\text{Fourier}}\mathcal{L}_{\text{Fourier}} \tag{8}$$

DiT Training Loss Once the VAE's training process is completed, its weights are frozen, and the DiT begins training. The DiT must learn to iteratively denoise corrupted latent representations while maintaining both the global artistic style and fine-grained details of the original artwork. The foundation of the diffusion loss function is its ability to predict the amount of noise added within each timestep of the forward diffusion process:

$$\mathcal{L}_{\text{MSE}}^{\text{noise}} = \|\epsilon - \hat{\epsilon}_\theta(z_t, t)\|^2 \tag{9}$$

where ϵ denotes the true noise, and $\hat{\epsilon}_\theta(z_t, t)$ represents the predicted noise at timestep t for the latent representation z_t. However, since the DiT operates within the latent space, the denoised latent representation must produce an equal or better reconstruction through the VAE decoder. To ensure the highest quality reconstructions, MSE (6), LPIPS, and Fourier Loss (7) are all used. By including the same three terms from the VAE's loss function, which strictly evaluate reconstruction accuracy, the proposed model maintains continuity and achieves more consistent results across the two stages. Furthermore, the hyperparameters β_{noise}, β_{img}, β_{LPIPS}, and β_{Fourier} are added to control the relative weighting of each loss component. The complete DiT training objective combines all of these components:

$$\mathcal{L}_{\text{DiT}} = \beta_{\text{noise}}\mathcal{L}_{\text{MSE}}^{\text{noise}} + \beta_{\text{img}}\mathcal{L}_{\text{MSE}}^{\text{img}} + \beta_{\text{LPIPS}}\mathcal{L}_{\text{LPIPS}} + \beta_{\text{Fourier}}\mathcal{L}_{\text{Fourier}} \tag{10}$$

4 Experiments

In this section, I present the artwork dataset used, describe the quantitative metrics to evaluate reconstructions, and demonstrate the performance of the proposed model through comprehensive comparisons with state-of-the-art methods and ablation studies.

4.1 Dataset and Metrics

Due to the large range of artworks in terms of artistic style and structure, I chose to use the "Best Artworks of All Time" dataset [8], publicly available on

Kaggle and curated by ikarus777, to create a model best suited for all cases. This dataset contains 8,446 images from 50 well-known artists, including Vincent van Gogh, Leonardo da Vinci, and Pablo Picasso, providing a broad representation of artistic styles and compositional structures. The diversity of the dataset makes it well-suited for learning generalized restoration patterns across different genres of visual art. Due to the unsupervised nature of the aforementioned vanilla models, no prompts or additional context will be provided during training or testing.

The models will be evaluated using three quantitative metrics to measure and validate their ability to accurately reconstruct the damaged artwork. Each evaluation metric aims to capture different aspects of the reconstruction's fidelity. L1 (Mean Absolute Error) measures pixel-wise differences, where 0 indicates perfect reconstruction:

$$\text{L1} = \frac{1}{N}\sum_{i=1}^{N}|x_i - \hat{x}_i| \tag{11}$$

where x_i is the original pixel values, $\hat{x}_i$ is the reconstructed pixel values, and N is the total number of pixels. PSNR (Peak Signal-to-Noise Ratio) measures overall fidelity, with values above 30 dB indicating high quality:

$$\text{PSNR} = 10 \cdot \log_{10}\left(\frac{\text{MAX}^2}{\text{MSE}}\right) \tag{12}$$

where MAX is the maximum pixel value and MSE is the mean squared error. SSIM (Structural Similarity Index) [23] evaluates perceptual similarity through luminance, contrast, and structure, with 1 indicating identical images:

$$\text{SSIM}(x, \hat{x}) = \frac{(2\mu_x\mu_{\hat{x}} + C_1)(2\sigma_{x\hat{x}} + C_2)}{(\mu_x^2 + \mu_{\hat{x}}^2 + C_1)(\sigma_x^2 + \sigma_{\hat{x}}^2 + C_2)} \tag{13}$$

where μ_x and $\mu_{\hat{x}}$ are the mean intensities, σ_x^2 and $\sigma_{\hat{x}}^2$ are the variances, $\sigma_{x\hat{x}}$ is the covariance, and C_1, C_2 are stability constants.

These three evaluation metrics will allow me to definitively determine the performance of the model with regard to the total accuracy, level of detail, and structural coherence throughout the reconstruction.

4.2 Experimental Results

Several state-of-the-art baseline models were used to evaluate the relative performance of the proposed model both qualitatively and quantitatively using the metrics above. These baseline models include: C3N [15], a content-constrained convolutional network using dual-domain partial convolution in spatial and frequency domains; PRN [26], a progressive reasoning network that decomposes restoration into luminance, sketch, and color reasoning subtasks; StrDiffusion [12], a structure-guided diffusion model that uses progressively sparse structural guidance to address semantic discrepancy; and ST-DiffNet [9], a diffusion-based network guided by structural and textural features with attention-based fusion modules. The reconstructed images from these models, including the proposed model,

were compared against the original image without the mask applied (ground truth).

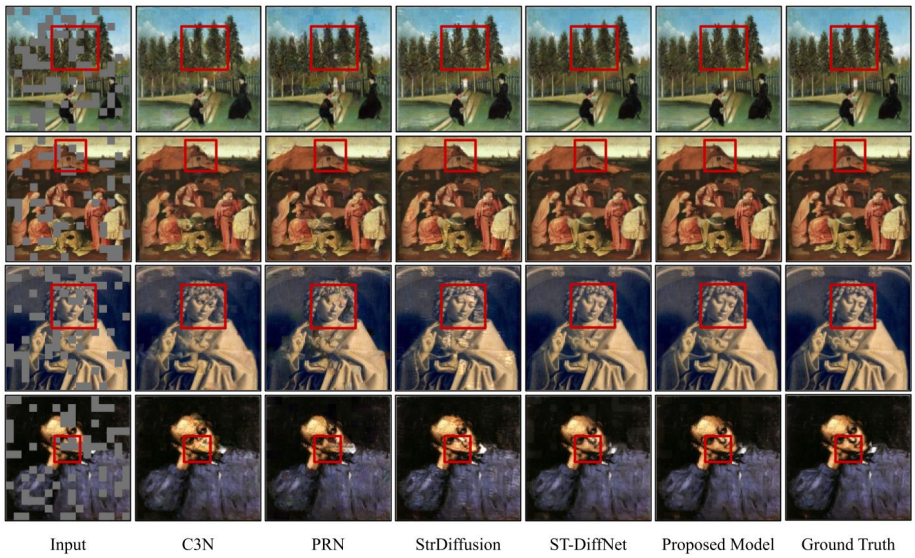


Fig. 3. Visual comparison of the reconstructions obtained by C3N [15], PRN [26], StrDiffusion [12], ST-DiffNet [9], and the proposed model from randomly selected images in the artwork dataset.

As illustrated by the highlighted regions in Figure 3, C3N [15] successfully leverages surrounding context to fill missing regions with appropriate hues, yet produces blurred edges and fails to recover fine facial details. PRN [26] achieves sharper edge definition but similarly struggles with facial feature reconstruction. StrDiffusion [12] excels at reconstructing both intricate background and facial details through its diffusion process, though it introduces a significant number of noisy artifacts. ST-DiffNet [9] achieves artifact-free reconstructions with better structural consistency, but tends to over-smooth fine details such as individual tree leaves (row 1) and facial contours present in the man's cheekbones (row 4). The proposed model addresses all these limitations, producing visually coherent reconstructions with minimal artifacts while preserving subtle color transitions and fine-grained details within the inpainted regions.

Table 1 presents the quantitative results comparing all baseline models against the proposed model. Notably, ST-DiffNet demonstrates strong performance with competitive PSNR (38.124 dB) and SSIM (0.934) scores, reflecting its effective use of structural and textural guidance through attention-based fusion modules. Nevertheless, the proposed model achieves superior performance across all metrics, with an 8.1% improvement in PSNR and 1.82% increase in SSIM compared

Table 1. Comparison of L1, PSNR, and SSIM [23] values from all models within the experiment. ↑ indicates that higher values are desired, ↓ indicates that lower values are desired. Bold text indicates the best values.

Model	L1 ↓	PSNR ↑	SSIM ↑
C3N	0.022	33.638	0.827
PRN	0.019	34.981	0.849
StrDiffusion	0.013	36.658	0.885
ST-DiffNet	0.012	38.124	0.934
Proposed Model	**0.010**	**41.212**	**0.951**

to ST-DiffNet, as well as a 16.67% reduction in L1 error. These improvements demonstrate the effectiveness of the proposed model as it produces both highly detailed and structurally unified reconstructions for damaged artwork.

The visual comparisons in Figure 3 alongside the quantitative metrics in Table 1 validate that the proposed model consistently outperforms existing methods in restoring diverse artwork across varying damaged areas.

4.3 Ablation Study

To ensure the necessity of each component of the proposed model, an ablation study was performed to compare the qualitative and quantitative performance of each part on the same artwork dataset. The ablation study consists of 3 baseline models alongside my proposed model: a vanilla VAE, a pixel-space diffusion model, and a Res-VAE [5]. The vanilla VAE consists of a standard encoder-decoder architecture with four up- and down-sampling blocks, using only basic convolution layers, batch normalization, and a ReLU activation function. The pixel-space diffusion model utilizes a modified DiT with cross-conditioning operating in pixel-space instead of latent space.

Figure 4 provides a visual comparison of ablation models against the proposed model given the masked input image. The Vanilla VAE successfully infers broad color regions from context but produces extremely blurry reconstructions that lack semantic coherence due to information loss at the architectural bottleneck. The Res-VAE demonstrates substantial improvement by accurately reconstructing general shapes, poses, and broad color distributions, yet fails to capture fine details such as the intricate folds in the scarf (row 1) and the nuanced cloud formations (row 2). The pixel-space diffusion model generates restorations with well-defined boundaries but often fails to blend them smoothly with the surrounding areas, causing visible boundary artifacts and texture inconsistencies. By operating in the latent space, the proposed model qualitatively outperforms all three baseline models to achieve both precise detail reconstruction and seamless integration with unmasked regions.

Table 2 presents the quantitative results comparing all ablation models against the proposed model. The proposed model achieves superior performance across all metrics, with a 33.3% reduction in L1 error and 11.5% improvement in PSNR compared to the Res-VAE, as well as a 2.59% increase in SSIM compared to the

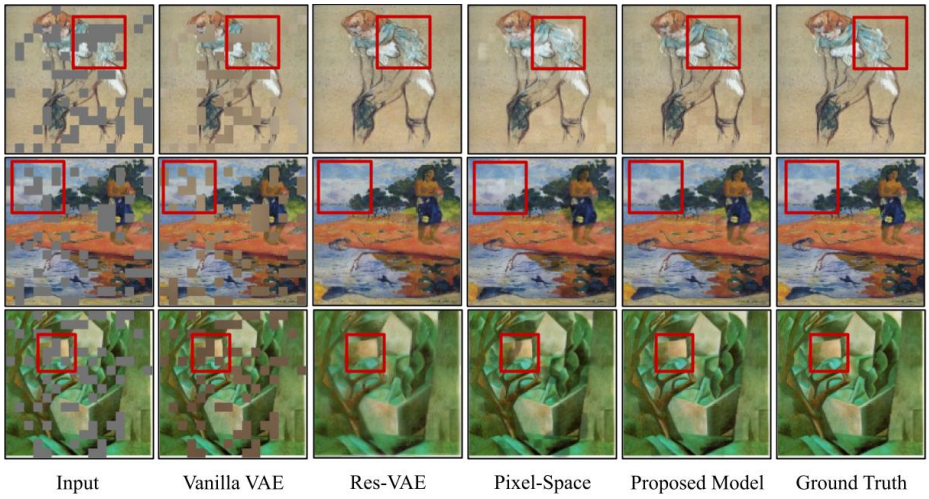


Fig. 4. Visual comparison of the reconstructions obtained by vanilla VAE, pixel-space diffusion model, Res-VAE [5], and the proposed model for randomly selected images from the artwork dataset.

Table 2. Comparison of L1, PSNR, and SSIM [23] values of the vanilla VAE, pixel-space diffusion model, Res-VAE [5], and the proposed model. ↑ indicates that higher values are desired, ↓ indicates that lower values are desired. Bold text indicates the best values.

Model	**L1 ↓**	**PSNR ↑**	**SSIM ↑**
Vanilla VAE	0.041	29.410	0.738
Res-VAE	0.015	36.959	0.889
Pixel-Space Diffusion	0.013	32.634	0.927
Proposed Model	**0.010**	**41.212**	**0.951**

pixel-space diffusion model. These improvements directly reflect the enhanced structural coherence and refined detail preservation achieved through iterative latent-space diffusion refinement.

Both Figure 4 and Table 2 demonstrate that the proposed model achieves better visual quality and quantitative metric values than the vanilla VAE, pixel-space diffusion model, and Res-VAE. This indicates that the proposed modules and methods are necessary and effective for artwork restoration.

4.4 Loss Ablation Study

To determine the optimal loss function configuration for VAE training, I conducted an ablation study comparing six configurations across 100 epochs with identical model parameters. I evaluated baseline models using only KL divergence and MSE, intermediate models incorporating LPIPS, and the full loss func-

tion (8). For each combination, I tested two weighting strategies: *equal* ($\beta = 1.0$ for all terms) and *optimized* (weights tuned through a grid search).

Table 3. Comparison of L1, PSNR, and SSIM [23] values for the Res-VAE reconstructions on six different loss function configurations. ↑ indicates that higher values are desired, ↓ indicates that lower values are desired. Bold text indicates the best values.

Loss Terms	Weight Type	L1 ↓	PSNR ↑	SSIM ↑
KL + MSE	Equal	0.0163	35.292	0.858
KL + MSE	Optimized	0.0166	35.394	0.859
KL + MSE + LPIPS	Equal	0.0163	35.716	0.869
KL + MSE + LPIPS	Optimized	0.0162	35.737	0.866
Full Loss Term	Equal	0.0154	36.267	0.878
Full Loss Term	Optimized	**0.0152**	**36.959**	**0.889**

As shown in Table 3, the baseline KL + MSE configuration achieves moderate performance with minimal improvement from optimization alone. Adding LPIPS improves structural quality (SSIM: 0.869) but produces mixed results when optimized. The full loss function demonstrates substantial improvements even with equal weighting, while the optimized configuration achieves the best results in L1, PSNR, and SSIM, reaching 0.0152 (-6.17%), 36.959 dB (+3.42%), and 0.889 (+2.66%), respectively, outperforming the optimized three-term loss function. These results validate the multi-faceted loss design where KL divergence and MSE provide foundational reconstruction, LPIPS adds perceptual guidance, and Fourier loss ensures fine detail preservation. The optimized hyperparameters ($\beta_{\text{KL}} = 0.001$, $\beta_{\text{MSE}} = 10$, $\beta_{\text{LPIPS}} = 100$, $\beta_{\text{Fourier}} = 1$) were used for all subsequent VAE and DiT training.

5 Conclusions

In this paper, I have proposed a novel latent diffusion method for high-fidelity artwork restoration across diverse time periods and artistic styles. By combining a Res-VAE with an image-conditioned diffusion transformer, the proposed model addresses fundamental limitations of existing artwork restoration approaches. The model leverages residual skip-connections for global and local feature consistency, compressed latent representations to reduce computational complexity, and transformer-based global attention to maintain stylistic coherence while gradually refining fine-grained details. The experimental results demonstrate that the proposed model achieves superior performance compared to state-of-the-art methods, with the ablation studies validating the necessity of each architectural component and the multi-faceted loss design. The proposed method advances the field of digital artwork restoration by providing a unified framework that handles diverse artistic styles without requiring manual tuning or additional context. Future research in artwork restoration should extend this latent diffusion approach to higher resolutions and varying mask patterns.

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